

WEATHERING, MASS MOVEMENT, GROUNDWATER / INDIA EROSION-LANDSLIDES-GROUNDWATER

EXOGENIC SOURCE-TRANSFER-STORAGE SYSTEM → WEATHERING TAXONOMY, CONTROLS, REGOLITH → SOIL EROSION SEQUENCE AND → MASS-MOVEMENT TAXONOMY AND SLOPE-STABILITY → INDIA LANDSLIDE GEOGRAPHY, RISK → GROUNDWATER ARCHITECTURE FROM RECHARGE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g10 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00

EXOGENIC SOURCE-TRANSFER-STORAGE SYSTEM

EXOGENIC SOURCE-TRANSFER-STORAGE SYSTEM

EXOGENIC LANDSCAPE SYSTEM

HOW IS ROCK CONVERTED INTO LANDFORM?

ROCK EXPOSED AT SURFACE WEATHERING IN PLACE REGOLITH AND

DENUDATION IS COLLECTIVE LOWERING; WEATHERING, EROSION AND MASS WASTING

MUST REMEMBER

WEATHERING PREPARES MATERIAL IN SITU, MASS MOVEMENT TRANSFERS IT

EXOGENIC LANDSCAPE SYSTEM

CENTRAL QUESTION: How is rock converted into landform?

ROCK EXPOSED AT SURFACE weathering in place regolith and loose material supplied gravity-driven transfer agent-driven

transfer mass wasting downslope water, wind, ice, waves erosion + transport deposition temporary storage remobilisation

Denudation is collective lowering; weathering, erosion and mass wasting are linked but distinct processes.

CORE CLAIM

- EXOGENIC LANDSCAPE SYSTEM
- CENTRAL QUESTION: How is rock converted into landform?
- ROCK EXPOSED AT SURFACE weathering in place regolith and loose material supplied gravity-driven transfer agent-driven

MECHANISM

- transfer mass wasting downslope water, wind, ice, waves erosion + transport deposition temporary storage remobilisation
- Denudation is collective lowering; weathering, erosion and mass wasting are linked but distinct processes.
- MUST REMEMBER: Weathering prepares material in situ, mass movement transfers it downslope, running water removes it, and groundwater

CONSEQUENCE / LIMIT

- Indian analysis links lithology, slope, rainfall, land use and pumping.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Denudation is a connected source–transfer–storage system, but weathering, erosion and mass wasting must still be distinguished by whether material remains in place, is carried by an agent, or moves mainly under gravity.

01

STAGE 01

WEATHERING TAXONOMY, CONTROLS, REGOLITH AND SOIL FORMATION

WEATHERING TAXONOMY, CONTROLS, REGOLITH AND SOIL FORMATION

MECHANICAL CHEMICAL BIOLOGICAL THERMAL SOLUTION ROOTS/BURROWS UNLOADING CARBONATION ORGANIC

REGOLITH ORGANIC INPUTS + ORGANISMS + TRANSLOCATION + TIME

R REMOVAL GAIN PARENT ROCK

CONTROLS CONVERGE

MINERALOGY + JOINTS/BEDDING + CLIMATE + RELIEF DRAINAGE +

BLACK SOIL LINK

DECCAN BASALT PARENT, SHRINK-SWELL CLAY, DEEP DRY CRACKS

CORE CLAIM

MECHANICAL CHEMICAL BIOLOGICAL thermal solution roots/burrows unloading carbonation organic acids frost hydration human exposure salt hydrolysis wet-dry oxidation
REGOLITH organic inputs + organisms + translocation + time

MECHANISM

O → A/E → B → C → R removal gain parent rock
CONTROLS CONVERGE: mineralogy + joints/bedding + climate + relief drainage + organisms + time + land use

CONSEQUENCE / LIMIT

Black soil link: Deccan basalt parent, shrink-swell clay, deep dry cracks
parent material does not alone determine every soil property.

CORE CLAIM

- MECHANICAL CHEMICAL BIOLOGICAL thermal solution roots/burrows unloading carbonation organic acids frost hydration human exposure salt hydrolysis wet-dry oxidation
- REGOLITH organic inputs + organisms + translocation + time

MECHANISM

- O → A/E → B → C → R removal gain parent rock
- CONTROLS CONVERGE: mineralogy + joints/bedding + climate + relief drainage + organisms + time + land use

CONSEQUENCE / LIMIT

- Black soil link: Deccan basalt parent, shrink-swell clay, deep dry cracks
- parent material does not alone determine every soil property.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Weathering is controlled not by climate alone but by the interaction of energy and moisture with mineralogy, discontinuities, relief, biota, time and land use.

02

STAGE 02

SOIL EROSION SEQUENCE AND WATERSHED CONTROL LOGIC

SOIL EROSION SEQUENCE AND WATERSHED CONTROL LOGIC

RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION DIFFUSE

CHAMBAL CUE

WATERSHED RESPONSE COVER SOIL

SLOW RUNOFF

SHORTEN SLOPE LENGTH

PROTECT DRAINAGE

TREAT GULLY HEAD

RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION diffuse sheet removal small rill channels entrenched

WATERSHED RESPONSE cover soil → slow runoff → shorten slope length → contour/terrace

protect drainage → treat gully head → rehabilitate catchment

WIND BRANCH surface creep → saltation → suspension

COASTAL BRANCH wave erosion belongs to the coastal-process system; use only when asked.

CORE CLAIM

- RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION diffuse sheet removal small rill channels entrenched gullies ravine landscape: Chambal cue
- WATERSHED RESPONSE cover soil → slow runoff → shorten slope length → contour/terrace

MECHANISM

- protect drainage → treat gully head → rehabilitate catchment
- WIND BRANCH surface creep → saltation → suspension

CONSEQUENCE / LIMIT

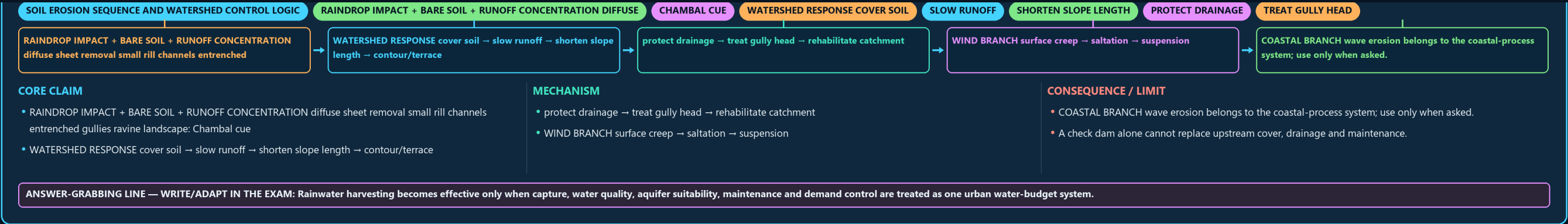
- COASTAL BRANCH wave erosion belongs to the coastal-process system; use only when asked.
- A check dam alone cannot replace upstream cover, drainage and maintenance.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Rainwater harvesting becomes effective only when capture, water quality, aquifer suitability, maintenance and demand control are treated as one urban water-budget system.

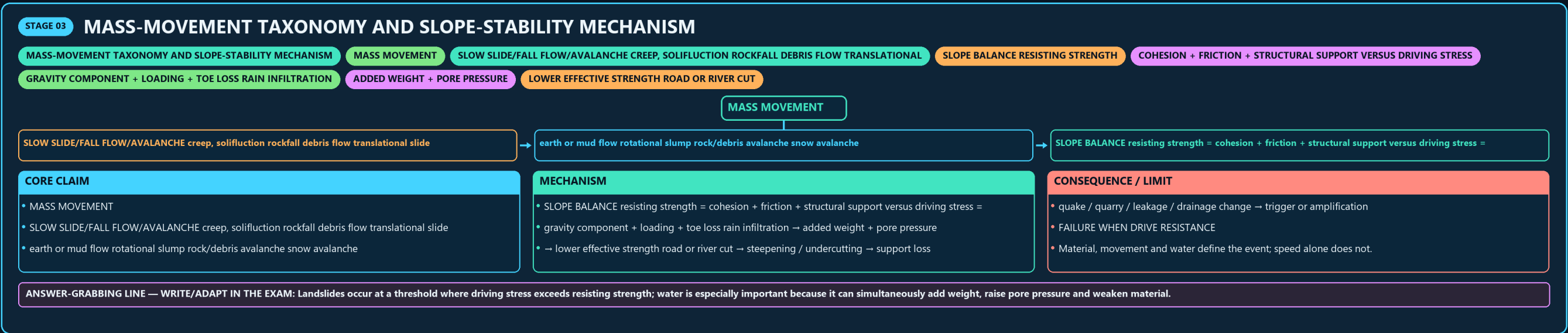
03

STAGE 03

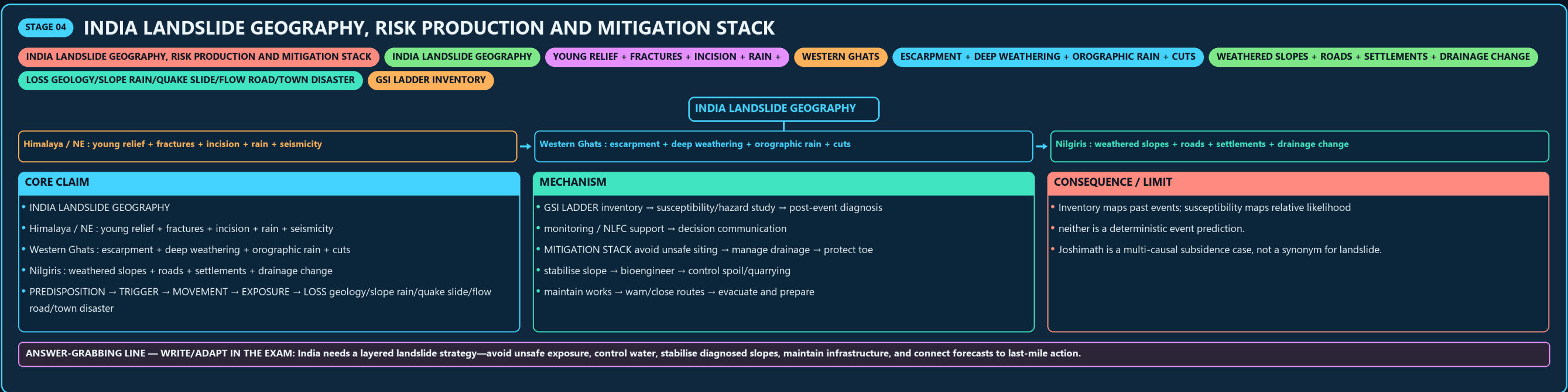
MASS-MOVEMENT TAXONOMY AND SLOPE-STABILITY MECHANISM



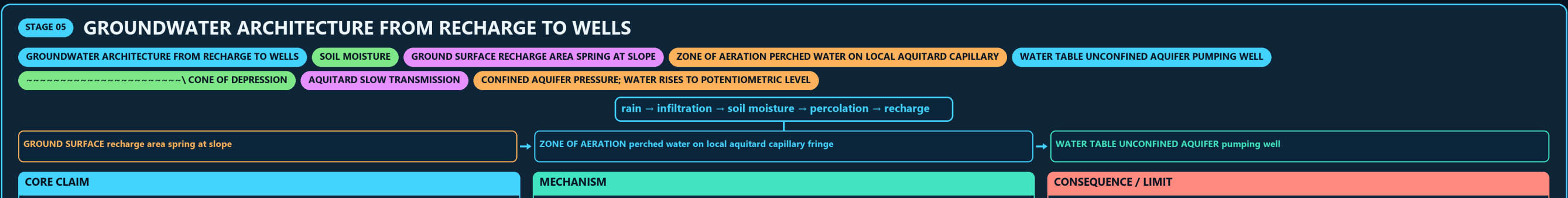
03



04



05



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: India needs a layered landslide strategy—avoid unsafe exposure, control water, stabilise diagnosed slopes, maintain infrastructure, and connect forecasts to last-mile action.

05

STAGE 05 GROUNDWATER ARCHITECTURE FROM RECHARGE TO WELLS

GROUNDWATER ARCHITECTURE FROM RECHARGE TO WELLS

SOIL MOISTURE

GROUND SURFACE RECHARGE AREA SPRING AT SLOPE

ZONE OF AERATION PERCHED WATER ON LOCAL AQUITARD CAPILLARY

WATER TABLE UNCONFINED AQUIFER PUMPING WELL

~~~~~\ CONE OF DEPRESSION

AQUITARD SLOW TRANSMISSION

CONFINED AQUIFER PRESSURE; WATER RISES TO POTENTIOMETRIC LEVEL

rain → infiltration → soil moisture → percolation → recharge

GROUND SURFACE recharge area spring at slope

ZONE OF AERATION perched water on local aquitard capillary fringe

WATER TABLE UNCONFINED AQUIFER pumping well

CORE CLAIM

- rain → infiltration → soil moisture → percolation → recharge
- GROUND SURFACE recharge area spring at slope
- ZONE OF AERATION perched water on local aquitard capillary fringe
- WATER TABLE UNCONFINED AQUIFER pumping well

MECHANISM

- ~~~~~\ cone of depression
- AQUITARD slow transmission
- CONFINED AQUIFER pressure; water rises to potentiometric level
- AQUICLUDE practically no transmission porosity = fraction of voids

CONSEQUENCE / LIMIT

- permeability = connected flow capacity aquifer stores and transmits
- aquitard transmits slowly artesian water rises under pressure; it flows only if head exceeds ground
- Budget: change in storage = recharge + inflow - discharge - pumping - outflow.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Groundwater is a dynamic stock governed by recharge, discharge, pumping, inter-aquifer flow and quality; annual recharge is therefore not a licence to abstract the same quantity everywhere.

06

STAGE 06 INDIA GROUNDWATER DEPLETION, QUALITY AND SUBSIDENCE GEOGRAPHY

INDIA GROUNDWATER DEPLETION, QUALITY AND SUBSIDENCE GEOGRAPHY

AQUIFER FORM

MAJOR PRESSURE

LAYERED ALLUVIUM

DEPLETION, AS, NITRATE

WEATHERED FRACTURES

DISCONTINUOUS STORAGE

DECCAN BASALT

REGION

Indo-Gangetic

Peninsular

Deccan basalt

Arid hard rock

Coastal belt

AQUIFER FORM

layered alluvium

weathered fractures

contacts + fractures

limited recharge

fresh-saline wedge

MAJOR PRESSURE

depletion, As, nitrate

discontinuous storage

uneven well yield

fluoride, salinity

intrusion and upconing crop-water demand + cheap energy + private wells + urban growth weak aquifer regulation + limited recharge falling head → lower yield → higher lift cost → unequal access compressible sediment; pore-pressure loss → compaction → subsidence

CORE CLAIM

- AQUIFER FORM
- MAJOR PRESSURE
- layered alluvium
- depletion, As, nitrate
- weathered fractures

MECHANISM

- contacts + fractures
- uneven well yield
- Arid hard rock
- limited recharge
- fluoride, salinity

CONSEQUENCE / LIMIT

- intrusion and upconing crop-water demand + cheap energy + private wells + urban growth
- weak aquifer regulation + limited recharge falling head → lower yield → higher lift
- cost → unequal access compressible sediment; pore-pressure loss → compaction → subsidence
- Quality paths differ: arsenic redox
- fluoride rock-water interaction nitrate sewage/fertiliser

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Good geography answers move from process to spatial pattern, then from consequence to a mechanism-matched and institutionally realistic response.

07

STAGE 07 RECHARGE, DEMAND GOVERNANCE AND INTEGRATED MAINS ANSWER SPINE

RECHARGE, DEMAND GOVERNANCE AND INTEGRATED MAINS ANSWER SPINE

GROUNDWATER SECURITY REQUIRES TWO CONVERGING ARMS

RECHARGE DEMAND

GOVERNANCE ROOF AND RUNOFF CAPTURE CROP-WATER ALIGNMENT MANAGED AQUIFER

AQUIFER PLAN

CGWB ASSESSMENT + NAQIM MAPPING + STATE REGULATION

ATAL BHUJAL LEARNING + LOCAL INSTITUTIONS + PUBLIC DATA

ANSWER SPINE

CORE CLAIM

- GROUNDWATER SECURITY REQUIRES TWO CONVERGING ARMS
- SUPPLY / RECHARGE DEMAND / GOVERNANCE roof and runoff capture crop-water alignment managed
- aquifer recharge efficient irrigation watershed treatment well registration/metering spring-shed restoration conjunctive use and
- reuse quality-safe source water community water budgets maintenance and monitoring enforceable state/local rules
- AQUIFER PLAN

MECHANISM

- Atal Bhujal learning + local institutions + public data
- ANSWER SPINE
- 1 define stock-flow-quality system
- 2 draw aquifer or India map
- 3 diagnose regional mechanism

CONSEQUENCE / LIMIT

- 5 match intervention to cause
- 6 qualify limits and equity
- Recharge structures are inputs, not proof of water-table recovery.
- Desalination supplies water but does not restore freshwater aquifer head.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Desalination supplies water but does not restore freshwater aquifer head.

RECHARGE, DEMAND GOVERNANCE AND INTEGRATED MAINS ANSWER SPINE

GROUNDWATER SECURITY REQUIRES TWO CONVERGING ARMS

RECHARGE DEMAND

GOVERNANCE ROOF AND RUNOFF CAPTURE CROP-WATER ALIGNMENT MANAGED AQUIFER

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ANSWER SPINE

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ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Desalination supplies water but does not restore freshwater aquifer head.

08

STAGE 08

EVIDENCE LADDER AND CONFIDENCE CONTROL

EVIDENCE LADDER AND CONFIDENCE CONTROL

EVIDENCE LADDER

FROM SOURCE BOOK

STANDARD KNOWLEDGE

CURRENT AFFAIRS.

2. AGRICULTURE.MD USES "CHERNOZEM", "ALLUVIAL" AND "BLACK SOIL" AS

3. EVIDENCE UNITS AVAILABLE IN THIS FILE

SOURCE TYPE CONTROLS THE STRENGTH AND LIMITS OF THE

CORE CLAIM

EVIDENCE LADDER

1. = from source book = inference / standard knowledge = current affairs.

MECHANISM

2. Agriculture.md uses "chernozem", "alluvial" and "black soil" as...

3. Evidence units available in this file

CONSEQUENCE / LIMIT

VERDICT: Source type controls the strength and limits of the historical claim.

EVIDENCE LADDER

CORE CLAIM

- EVIDENCE LADDER
- 1. = from source book = inference / standard knowledge = current affairs.

MECHANISM

- 2. Agriculture.md uses "chernozem", "alluvial" and "black soil" as...
- 3. Evidence units available in this file

CONSEQUENCE / LIMIT

- VERDICT: Source type controls the strength and limits of the historical claim.
- EVIDENCE LADDER

MECHANISM / VERDICT — VERDICT: Source type controls the strength and limits of the historical claim.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Source type controls the strength and limits of the historical claim.

09

STAGE 09

EXAMINER TRAPS AND CONTESTED BOUNDARIES

EXAMINER TRAPS AND CONTESTED BOUNDARIES

CLOSE DISTINCTIONS

1. CAUSAL METHOD INITIAL CONDITION

ENERGY/FORCE OR DRIVER →

2. SCALE, INTERACTING CONTROL OR EVIDENCE LIMIT

3. UPSC TRAP

A LANDSLIDE CAN DELIVER SEDIMENT TO A RIVER

QUALIFICATION EARNS MARKS; FALSE CERTAINTY IS A HARD FAILURE.

CORE CLAIM

CLOSE DISTINCTIONS

1. Causal method Initial condition → energy/force or driver →...

2. SCALE, INTERACTING CONTROL OR EVIDENCE LIMIT

MECHANISM

3. UPSC TRAP / ANSWER-USE: A landslide can deliver sediment to a river,...

VERDICT: Qualification earns marks; false certainty is a hard failure.

CLOSE DISTINCTION: Weathering differs from erosion, porosity from permeability, water table from piezometric surface,

CONSEQUENCE / LIMIT

aquifer from aquitard, and slump/slide/flow/creep require movement-specific geometry rather than the generic label landslide.

CORE CLAIM

- CLOSE DISTINCTIONS
- 1. Causal method Initial condition → energy/force or driver →...
- 2. SCALE, INTERACTING CONTROL OR EVIDENCE LIMIT

MECHANISM

- 3. UPSC TRAP / ANSWER-USE: A landslide can deliver sediment to a river,...
- VERDICT: Qualification earns marks; false certainty is a hard failure.
- CLOSE DISTINCTION: Weathering differs from erosion, porosity from permeability, water table from piezometric surface,

CONSEQUENCE / LIMIT

- aquifer from aquitard, and slump/slide/flow/creep require movement-specific geometry rather than the generic label landslide.

MECHANISM / VERDICT — VERDICT: Qualification earns marks; false certainty is a hard failure.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Qualification earns marks; false certainty is a hard failure.

10

STAGE 10

INTEGRATED ANSWER SPINE AND QUALIFIED CONCLUSION

INTEGRATED ANSWER SPINE AND QUALIFIED CONCLUSION

ANSWER SPINE

1. ORIGINAL SOLVED MAINS QUESTIONS 6

2. LOCAL OFFICIAL PAPERS/KEYS CHECKED

2018, 2019, 2020, 2021, 2023

3. PRACTICE CONTRACT EVERY SOLVED ITEM HAS DEMAND DECODING

CLAIM, NAMED EVIDENCE, ANALYSIS AND QUALIFICATION LEAD TO A

EVIDENCE LIMIT

ANSWER SPINE



EROSION, INTERACTING CONTROL ON EVIDENCE LIMIT

EROSION, INTERACTING CONTROL ON EVIDENCE LIMIT

MECHANISM / VERDICT — VERDICT: Qualification earns marks; false certainty is a hard failure.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Qualification earns marks; false certainty is a hard failure.

10

STAGE 10

INTEGRATED ANSWER SPINE AND QUALIFIED CONCLUSION

INTEGRATED ANSWER SPINE AND QUALIFIED CONCLUSION

ANSWER SPINE

1. ORIGINAL SOLVED MAINS QUESTIONS 6

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2018, 2019, 2020, 2021, 2023

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CLAIM, NAMED EVIDENCE, ANALYSIS AND QUALIFICATION LEAD TO A

EVIDENCE LIMIT

ANSWER SPINE

1. Original solved Mains questions 6

2. Local official papers/keys checked: 2018, 2019, 2020, 2021, 2023

3. Practice contract Every solved item has demand decoding, a detailed

CORE CLAIM

• ANSWER SPINE

• 1. Original solved Mains questions 6

• 2. Local official papers/keys checked: 2018, 2019, 2020, 2021, 2023,...

MECHANISM

• 3. Practice contract Every solved item has demand decoding, a detailed...

• VERDICT: Claim, named evidence, analysis and qualification lead to a graded...

• EVIDENCE LIMIT: A trigger such as intense rain is not the complete cause: antecedent moisture, discontinuities, toe cutting,

CONSEQUENCE / LIMIT

• groundwater trends and notified areas require dated official status.

MECHANISM / VERDICT — VERDICT: Claim, named evidence, analysis and qualification lead to a graded...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Claim, named evidence, analysis and qualification lead to a graded.

11

STAGE 11

EVIDENCE LADDER AND CONFIDENCE CONTROL — DEEP-REVIEW SYNTHESIS 2

EVIDENCE LADDER AND CONFIDENCE CONTROL — DEEP-REVIEW SYNTHESIS 2

EVIDENCE LADDER

FROM SOURCE BOOK

STANDARD KNOWLEDGE

CURRENT AFFAIRS.

2. AGRICULTURE.MD USES "CHERNOZEM", "ALLUVIAL" AND "BLACK SOIL" AS

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EVIDENCE LADDER

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2. Agriculture.md uses "chernozem", "alluvial" and "black soil" as

3. Evidence units available in this file

VERDICT: Source type controls the strength and limits of the historical claim.

CORE CLAIM

• EVIDENCE LADDER

• 1. = from source book = inference / standard knowledge = current affairs.

MECHANISM

• 2. Agriculture.md uses "chernozem", "alluvial" and "black soil" as...

• 3. Evidence units available in this file

CONSEQUENCE / LIMIT

• VERDICT: Source type controls the strength and limits of the historical claim.

• EVIDENCE LADDER

MECHANISM / VERDICT — VERDICT: Source type controls the strength and limits of the historical claim.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Pedogenesis begins with weathered parent material but becomes soil formation only when organic activity, translocation and time create an organised profile; black soil illustrates how basaltic parent material and seasonal moisture regimes shape properties.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

ADVANCED 1 — EFFECTIVE STRESS, FACTOR OF SAFETY AND

ADVANCED 2 — WEATHERING PROFILES, CLAYS AND HYDRO-MECHANICAL FEEDBACK

ADVANCED 3 — FORECAST SKILL, FALSE ALARMS AND NON-STATIONARITY

ADVANCED 4 — AQUIFER HETEROGENEITY, CAPTURE AND SUSTAINABLE YIELD

ADVANCED 5 — GOVERNANCE EVALUATION

IN TECHNICAL TERMS, SHEAR STRENGTH IS OFTEN REPRESENTED BY

COHESION PLUS EFFECTIVE NORMAL STRESS MULTIPLIED BY FRICTION.

OPTIONAL SOURCE DEPTH

• ADVANCED 1 — EFFECTIVE STRESS, FACTOR OF SAFETY AND THRESHOLDS

• ADVANCED 2 — WEATHERING PROFILES, CLAYS AND HYDRO-MECHANICAL FEEDBACK

• ADVANCED 3 — FORECAST SKILL, FALSE ALARMS AND NON-STATIONARITY

• ADVANCED 4 — AQUIFER HETEROGENEITY, CAPTURE AND SUSTAINABLE YIELD

ADVANCED DEBATE

• ADVANCED 5 — GOVERNANCE EVALUATION

• ADVANCED 1 — EFFECTIVE STRESS, FACTOR OF SAFETY AND THRESHOLDS The simple core rule is driving stress versus resisting strength.

• In technical terms, shear strength is often represented by a Mohr–Coulomb relation: cohesion plus effective normal stress multiplied by friction.

• Pore pressure lowers effective normal stress.

LEGACY / NUANCE

• A factor of safety above one indicates calculated resistance exceeds driving force; near one means marginal stability.

• Real slopes remain uncertain because geometry, groundwater pathways and material properties vary.

• ADVANCED 2 — WEATHERING PROFILES, CLAYS AND HYDRO-MECHANICAL FEEDBACK Weathering fronts can create saprolite, corestones, clay-rich horizons and permeability contrasts.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.